

AstroMLab 4: Benchmark-Topping Performance in Astronomy Q&A with a 70B-Parameter Domain-Specialized Reasoning Model

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General-purpose large language models (LLMs), despite their broad capabilities, often struggle with specialized domain knowledge, a limitation particularly pronounced in more accessible, lower-parameter versions. This gap hinders their deployment as effective agents in demanding fields such as astronomy. Building on our prior work with AstroSage-Llama-3.1-8B, this study introduces AstroSage-Llama-3.1-70B, a significantly larger and more advanced domain-specialized natural-language AI assistant. It is designed for research and education across astronomy, astrophysics, space science, astroparticle physics, cosmology, and astronomical instrumentation. Developed from the Meta-Llama-3.1-70B foundation, AstroSage-Llama-3.1-70B underwent extensive continued pre-training (CPT) on a vast corpus of astronomical literature, followed by supervised fine-tuning (SFT) and model merging. Beyond its 70-billion parameter scale, this model incorporates refined datasets, judiciously chosen learning hyperparameters, and improved training procedures, achieving state-of-the-art performance on complex astronomical tasks. Notably, we integrated reasoning chains into the SFT dataset, enabling AstroSage-Llama-3.1-70B to either answer the user query immediately, or first emit a human-readable thought process. Evaluated on the AstroMLab-1 benchmark [1]—comprising 4,425 questions from literature withheld during training—AstroSage-Llama-3.1-70B achieves state-of-the-art performance. It surpasses all other tested open-weight and proprietary models, including leading systems like o3, Gemini-2.5-Pro, Claude-3.7-Sonnet, Deepseek-R1, and Qwen-3-235B, even those with API costs two orders of magnitude higher. This work demonstrates that domain specialization, when applied to large-scale models, can enable them to outperform generalist counterparts in specialized knowledge areas like astronomy, thereby advancing the frontier of AI capabilities in the field.

Keywords: AI assistant, large-language model, continued pretraining, supervised fine-tuning, model merging, astronomy, astrophysics, cosmology

I. INTRODUCTION

Astronomy and its related fields demand sophisticated tools that can process vast amounts of specialized knowledge. Large Language Models (LLMs) have emerged as promising assistants for this domain, offering capabilities as research collaborators, educational resources, and knowledge repositories. Domain-specialized models demonstrate particular cost-effectiveness in such contexts, as their parameters can be optimized for specific knowledge domains rather than distributed across the breadth of general internet content. Our previous work, AstroSage-Llama-3.1-8B [2] (hereafter TdH25), established that a relatively modest 8-billion parameter LLM, when extensively trained on astronomical content, could match or exceed the performance of much larger

general-purpose models on astronomical knowledge tasks [1]. This finding highlighted the potential of domain specialization for creating efficient, high-performing AI assistants.

We now introduce AstroSage-Llama-3.1-70B, a 70-billion parameter language model that represents an advancement in specialized AI for astronomy. Our central research question asks whether domain specialization merely improves efficiency or can enable specialized models to outperform even the largest commercial alternatives. Following TdH25’s approach with Meta-Llama-3.1-8B, we applied similar domain specialization techniques to the larger Meta-Llama-3.1-70B foundation. Beyond the increased parameter count, we implemented several key enhancements: expanded and refined datasets for both continued pre-training (CPT) and supervised

fine-tuning (SFT); optimized learning hyperparameters based on public benchmarks and our own experimentation; and an explicit reasoning capability that enables step-by-step analytical processes before generating answers.

The core hypothesis driving this work is that a larger specialized model can elevate AI performance across astronomy, astrophysics, space science, astroparticle physics, cosmology, and astronomical instrumentation—collectively referred to as “astronomy” throughout this paper. While AstroSage-Llama-3.1-8B successfully matched larger models’ performance, AstroSage-Llama-3.1-70B aims to surpass even advanced commercial alternatives. Beyond testing this hypothesis, we are making our trained model openly available to serve as a resource for researchers, educators, and students in the field, as well as non-scientists interested in astronomy.

This paper is structured as follows: Section II details the multi-stage training pipeline for AstroSage-Llama-3.1-70B, including improvements to our CPT and SFT corpora and procedures, and the model merging strategy for the final release. Section III discusses potential applications. Section IV evaluates AstroSage-Llama-3.1-70B against the updated Ting *et al.* [1] benchmark, comparing it with other recent proprietary and open-weight models.

II. MODEL ARCHITECTURE AND TRAINING METHODOLOGY

AstroSage-Llama-3.1-70B is derived from the Meta-Llama-3.1-70B architecture [3]. This base model was selected for consistency with TdH25, which chose Meta-Llama-3.1-8B for its state-of-the-art general capabilities and permissive licensing. The tokenizer from Meta-Llama-3.1-70B-Instruct was used without modification. Following the methodology of TdH25, our development process comprised three main stages: continued pre-training (CPT), supervised fine-tuning (SFT), and model merging.

A. Continued Pre-training (CPT)

The objective of CPT is to imbue the base model with extensive domain-specific knowledge from astronomical literature. The CPT dataset for AstroSage-Llama-3.1-70B builds upon the comprehensive corpus developed in TdH25, which included approximately 250,000 arXiv preprints (astro-ph, gr-qc from 2007-2024), nearly 30,000 Wikipedia articles related to astronomy, and internet-available textbooks. The knowledge cutoff for the astroph papers included in the CPT dataset remains January 2024.

This dataset was further enhanced:

- We applied FTFY [4] for consistent Unicode text normalization and some rule-based repetition removal to correct OCR failures, supplementing

the perplexity-based cleaning methods detailed in TdH25.

- To preserve general language understanding and mitigate catastrophic forgetting due to specialization, we incorporated a random selection of samples from the FineWeb dataset [5] into each training epoch. This addition of previous pretraining tokens during CPT is sometimes known as “replay”. Notably, the specific FineWeb samples were varied for each epoch, ensuring diverse exposure to general web text.

The CPT phase was conducted on the Oak Ridge Leadership Computing Facility (OLCF) Frontier supercomputer utilizing AMD Instinct MI250X GPUs. The Meta-Llama-3.1-70B base model underwent 2.5 epochs of CPT, an extensive training process that utilized approximately 168,000 GPU-hours. Several learning hyperparameters were modified compared to TdH25. Firstly, the learning rate was scaled relative to that in Dubey *et al.* [3] to maintain the same *learning rate per token*. Secondly, the warm-up period was extended to 0.15 epochs. Finally, we increased the weight decay regularization term to 0.1, aligning with Dubey *et al.* [3].

B. Supervised Fine-tuning (SFT)

Following CPT, the model underwent SFT to develop its instruction-following and conversational capabilities, and to instill behaviors such as chain-of-thought and self-reflection (collectively termed “reasoning”).

Figure 1 illustrates the composition of the SFT dataset. Its largest component is NVIDIA’s Llama-Nemotron-Post-Training-Dataset [6], which was used to train the Llama-3.1-Nemotron series. This series consistently demonstrates excellent performance on the LMArena [7], suggesting it is a strong dataset for eliciting reasoning and aligning with human preferences. The reasoning components cover science, code, mathematics, and general chat, providing a foundation for analytical thinking across different domains.

We also included the OpenHermes 2.5 dataset, which helps build general instruction-following capabilities and adherence to the system prompt.

To enhance domain expertise, we incorporated the custom domain-specific Q&A datasets from both TdH25 and de Haan [8], which together comprise approximately 30% of the training data.

After combination, the dataset was deduplicated and shuffled. A loss mask was applied to train the model exclusively on assistant completions, excluding user queries and system prompts. The chat template adheres to the Llama-3.1 standard.

This diverse SFT mixture aims to produce a model that is knowledgeable in astronomy, has reasoning ability, and is good at following instructions.

Supervised Fine-tuning Data by JSONL File Size (MB)

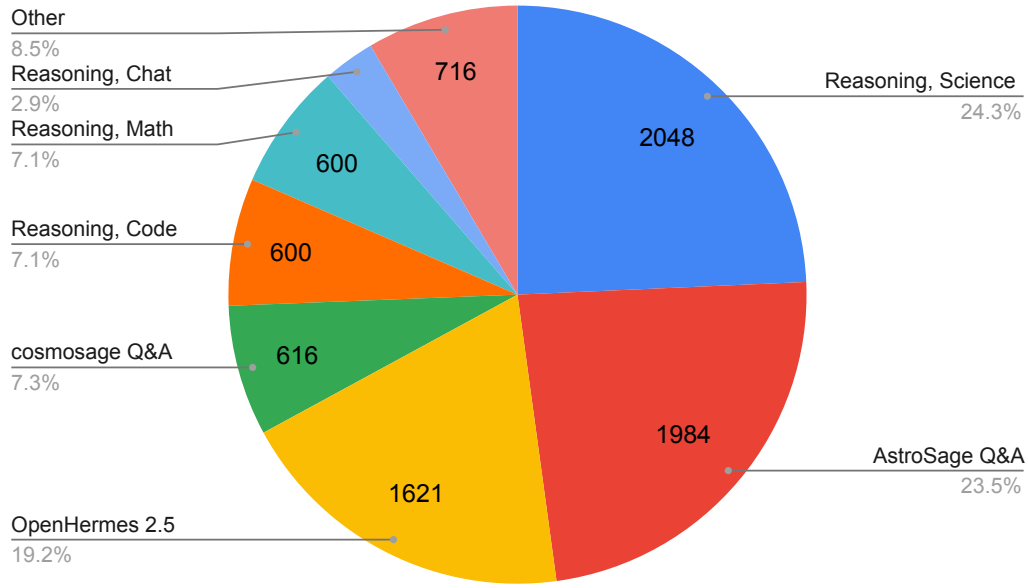


FIG. 1. Composition of the AstroSage-Llama-3.1-70B SFT training dataset. The combination of reasoning-focused datasets (41.8%) with domain-specific astronomy Q&A (30.8%) reflects our strategy to develop a model that combines analytical thinking with specialized knowledge. General instruction-following data (OpenHermes 2.5) helps maintain versatility while preserving domain expertise. File sizes represent uncompressed UTF-8 encoded text including system and user prompts.

The model was fine-tuned on this SFT dataset for 0.6 epochs. This phase consumed approximately 13,000 GPU-hours on the same MI250X GPU architecture. Hyperparameters mirrored those of the CPT stage, with the exception of weight decay, which was removed.

C. Training Implementation

The CPT and SFT stages were conducted on the Oak Ridge Leadership Computing Facility (OLCF) Frontier supercomputer using AMD Instinct MI250X GPUs. Our implementation employed the GPT-NeoX framework [9], which we adapted for compatibility with the Llama-3.1 architecture. Training was distributed across 2,048 Graphics Compute Dies (GCDs) using a multi-dimensional parallelism strategy: tensor parallelism 8, pipeline parallelism 8, and data parallelism 32. As GPT-NeoX does not currently support DeepSpeed ZeRO stage 2/3 with pipeline parallelism, we used ZeRO stage 1 [10] with activation checkpointing enabled. This configuration achieved a computational throughput of approximately 50 TFLOPS/s per GCD, consistent with performance metrics reported by Dash *et al.* [11].

Figure 2 illustrates the training dynamics during both CPT and SFT phases. Several noteworthy patterns emerge from these curves. The CPT loss exhibits a consistent downward trajectory without discontinuities at

epoch boundaries—a positive indicator that the model was still learning effectively from the dataset without overfitting. This suggests that additional training epochs would likely have yielded further performance improvements, had computational resources permitted.

Resource constraints limited our training to 2.5 epochs for CPT and 0.6 epochs for SFT, resulting in an incomplete final annealing stage and preventing full convergence of the SFT phase. Despite these limitations, AstroSage-Llama-3.1-70B demonstrates superior performance in astronomy Q&A benchmarks compared to other leading models. With additional compute resources, we anticipate further performance gains as the training could be extended to leverage the full potential of our dataset.

The SFT training loss initially exceeds the final CPT loss but rapidly decreases, ultimately dropping below 0.6. This indicates that, on average, the model assigns more than half its probability ($\langle P \rangle = \exp(-\text{loss})$) to the correct token—likely reflecting the more predictable nature of assistant responses compared to the CPT corpus. The model obtained after SFT is named AstroSage-70B-SFT.

D. Model Merging

To create the final, publicly released AstroSage-Llama-3.1-70B, we employed model merging (also known as pa-

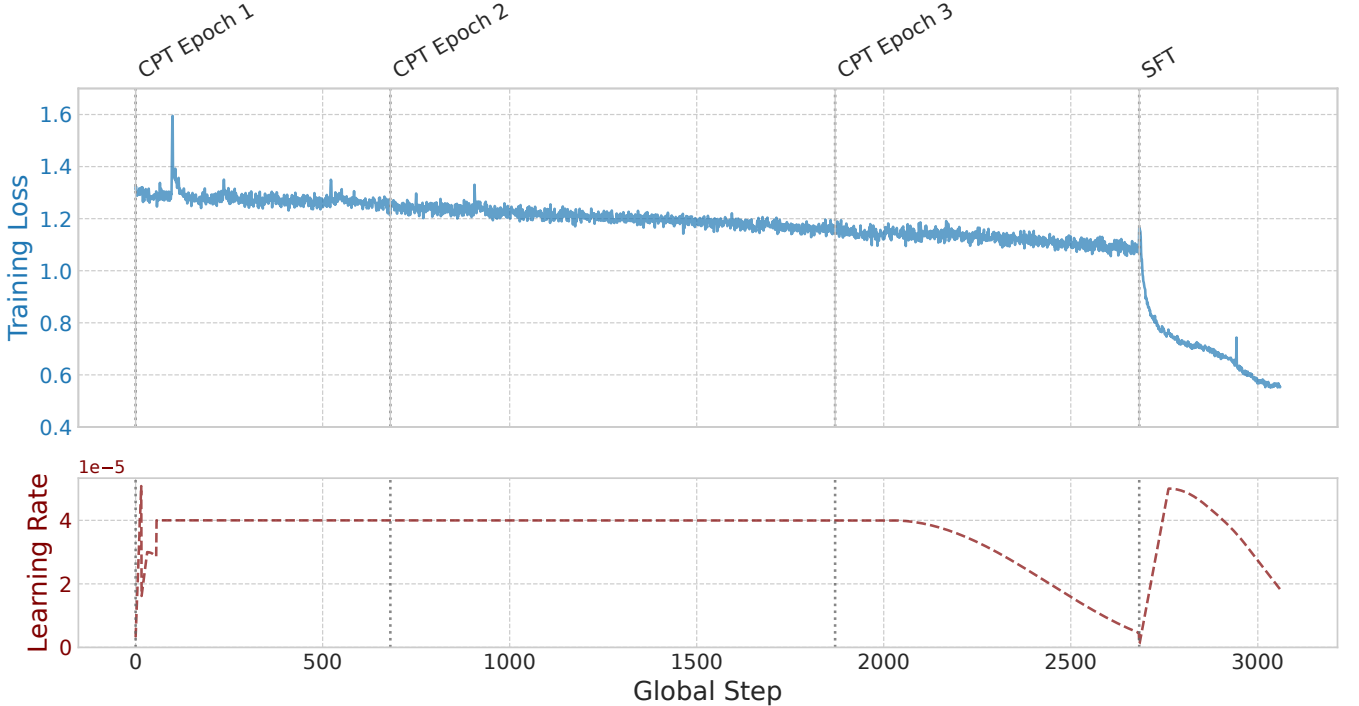


FIG. 2. Training dynamics for continued pre-training (CPT) and supervised fine-tuning (SFT). The top panel shows the training loss trajectory across 2.5 epochs of CPT followed by 0.6 epochs of SFT. Despite an early significant spike during CPT Epoch 1, the loss steadily decreases throughout training with minimal discontinuity at epoch boundaries, indicating effective learning without overfitting. The SFT phase exhibits a rapid initial decrease in loss, eventually reaching below 0.6. The bottom panel reveals the learning rate schedule, including the initial warm-up period, several manual adjustments during early CPT (visible as step changes), the planned cosine decay during the final partial CPT epoch, and a separate warm-up and decay cycle for the SFT phase. Early termination of both phases reflects the computational resource constraints.

parameter averaging) using the `mergekit` library [12]. This technique allows us to combine the strengths of our specialized SFT model with the robust instruction-following capabilities of other popular fine-tuned Llama-3.1-70B variants.

The final mixture for AstroSage-Llama-3.1-70B was created using the DARE-TIES method [13] with parameters `rescale: true` and `lambda: 1.2`. The AstroSage-70B-CPT model (our domain-specialized, pre-instruction-tuning model) was designated as the “base model” for the merge operation. The components and their contributions are:

- 70% AstroSage-70B-SFT (density 0.7)
- 15% Llama-3.1-Nemotron-70B-Instruct (density 0.5)
- 7.5% Meta-Llama-3.3-70B-Instruct (density 0.5)
- 7.5% Meta-Llama-3.1-70B-Instruct (density 0.5)

This model mixture and its associated hyperparameters were determined through an iterative trial-and-error process, aiming to maximize performance on an internal set of astronomy-focused questions, distinct from the benchmark we use in Section IV. Consequently, the final model

inherits the astronomical expertise from AstroSage-70B-SFT while incorporating the refined general instruction-following and conversational polish from the other constituent models.

III. FEATURES AND REASONING CAPABILITY

AstroSage-Llama-3.1-70B is designed for a wide range of applications within the astronomical domain, offering capabilities similar to AstroSage-Llama-3.1-8B but enhanced by its larger scale and refined training. Potential applications include addressing factual queries (verifiable or low-risk), literature review, summarization, assisting with manuscript preparation and revision, brainstorming, hypothesis formulation, concept learning, programming support, and as a component in an agentic system.

A notable feature of AstroSage-Llama-3.1-70B is its explicit reasoning capability. This aligns with recent advances in the broader LLM field, where explicit reasoning has emerged as a critical development for enhancing model performance on complex tasks. The integration of reasoning mechanisms has become increasingly common in state-of-the-art models, including OpenAI’s

o1 through o4 series, Anthropic’s Claude models with “thinking mode,” DeepSeek-R1, and others. These developments demonstrate that exposing and structuring the internal reasoning process allows models to tackle complex problems more systematically, resulting in improved accuracy and reliability.

Building on these industry-wide insights, AstroSage-Llama-3.1-70B implements explicit reasoning that can be activated at inference time by setting the system prompt to `detailed thinking on` and prefilling the assistant completion with `<think>`. When enabled, the model generates a step-by-step reasoning process before providing the final answer to the user’s query. This is particularly beneficial for complex astronomical problems requiring multi-step analysis. As the reasoning tokens are enclosed within `<think>` and `</think>` tags, they can easily be hidden from the end-user if desired.

IV. EVALUATION

To evaluate the performance of AstroSage-Llama-3.1-70B, we utilize the AstroMLab-1 benchmark developed in Ting *et al.* [1]. This benchmark consists of 4,425 high-quality, human-verified multiple-choice questions spanning astronomy, astrophysics, cosmology, and astronomical instrumentation. Importantly, these questions are derived from Annual Review of Astronomy and Astrophysics (ARAA) papers that were explicitly withheld from the AstroSage training corpus. This ensures the model is evaluated on genuinely unseen material, and its performance is not merely an artifact of training on the benchmark’s source texts.

On this benchmark, AstroSage-Llama-3.1-70B achieves a score of 86.2% without enabling reasoning. As illustrated in Figure 3, this performance establishes AstroSage-Llama-3.1-70B as the leading model, outperforming all other tested open-weight and proprietary models. This includes a substantial improvement over its 8-billion parameter predecessor, AstroSage-Llama-3.1-8B, as well as contemporary large-scale general-purpose LLMs such as o3, GPT-4.1, Claude-3.7-Sonnet, and Deepseek-R1. For context, as mentioned in TdH25, professional astronomers score around 67% on this benchmark.

Our evaluation substantially updates the results originally presented in Ting *et al.* [1], which was published in July 2024. The benchmark analysis includes a cost-accuracy trade-off visualization, represented by diagonal dashed lines in Figure 3. This analysis reveals that within a model family (e.g., GPT-4.1 series or Gemini series), a tenfold increase in API cost typically corresponds to an improvement of approximately 3.5 percentage points in accuracy. Consequently, the distance between adjacent lines represents an order of magnitude gain in cost efficiency.

As highlighted by the vertical red arrows in Figure 3, our domain specialization approach delivers remarkable

efficiency gains. Both AstroSage models jump approximately two cost-efficiency lines compared to their respective base models (Llama-3.1-8B and Llama-3.1-70B), representing improvement of approximately $100\times$ in cost-efficiency. This demonstrates that targeted domain specialization can achieve performance levels that would otherwise require models costing two orders of magnitude more at inference time.

For testing, we provided all models with minimal system prompts indicating they should act as astronomy experts. This approach ensures homogeneous evaluation conditions across all models. We verified that for the high-performing models included in our comparison, variables such as temperature settings played a relatively minor role in performance outcomes, further confirming the robustness of our comparative analysis.

The original study predicted model improvements would shift performance to the next diagonal line every three to six months, a forecast that has proven accurate. Most leading models from July 2024 performed at the level where Step-2¹ now resides, and we observe that current models have largely surpassed this efficiency level. A more detailed analysis of these trends will be presented in a forthcoming paper dedicated to benchmarking efforts.

An interesting observation from our updated evaluation is that while the 3.5 percentage point trade-off slope still holds for some series (notably the Gemini-2.0 and Gemini-2.5 series), models like Qwen-3 and GPT-4.1 exhibit steeper drop-offs in performance across their tiered offerings. This suggests that current distillation approaches for creating more affordable variants of powerful models may be less effective for specialized knowledge domains like astronomy.

For our cost analysis, we took the average of the cost of input tokens and the cost of output tokens, as listed at <https://artificialanalysis.ai/leaderboards/providers> at the time of testing. While these costs may fluctuate by tens of percentage points, such variations would not significantly alter the comparative landscape presented in Figure 3.

In our evaluation methodology, we applied a consistent approach to models with reasoning capabilities. For models with explicit reasoning modes (e.g., Gemini-2.5-Flash), we enabled this feature during testing. While this approach may not yield perfectly equivalent cost comparisons due to the increased token usage in reasoning models, it provides a standardized basis for performance assessment. Interestingly, we found that enabling reasoning modes generally did not significantly improve scores for most models on this benchmark, including AstroSage-Llama-3.1-70B. This finding may be due to the AstroMLab-1 questions primarily testing System 1 thinking (fast, intuitive knowledge recall) rather

¹ The Step-2 results presented here reflect the version of the model as of November 2024, which differs from the version evaluated in the original Ting *et al.* [1] study in July 2024.

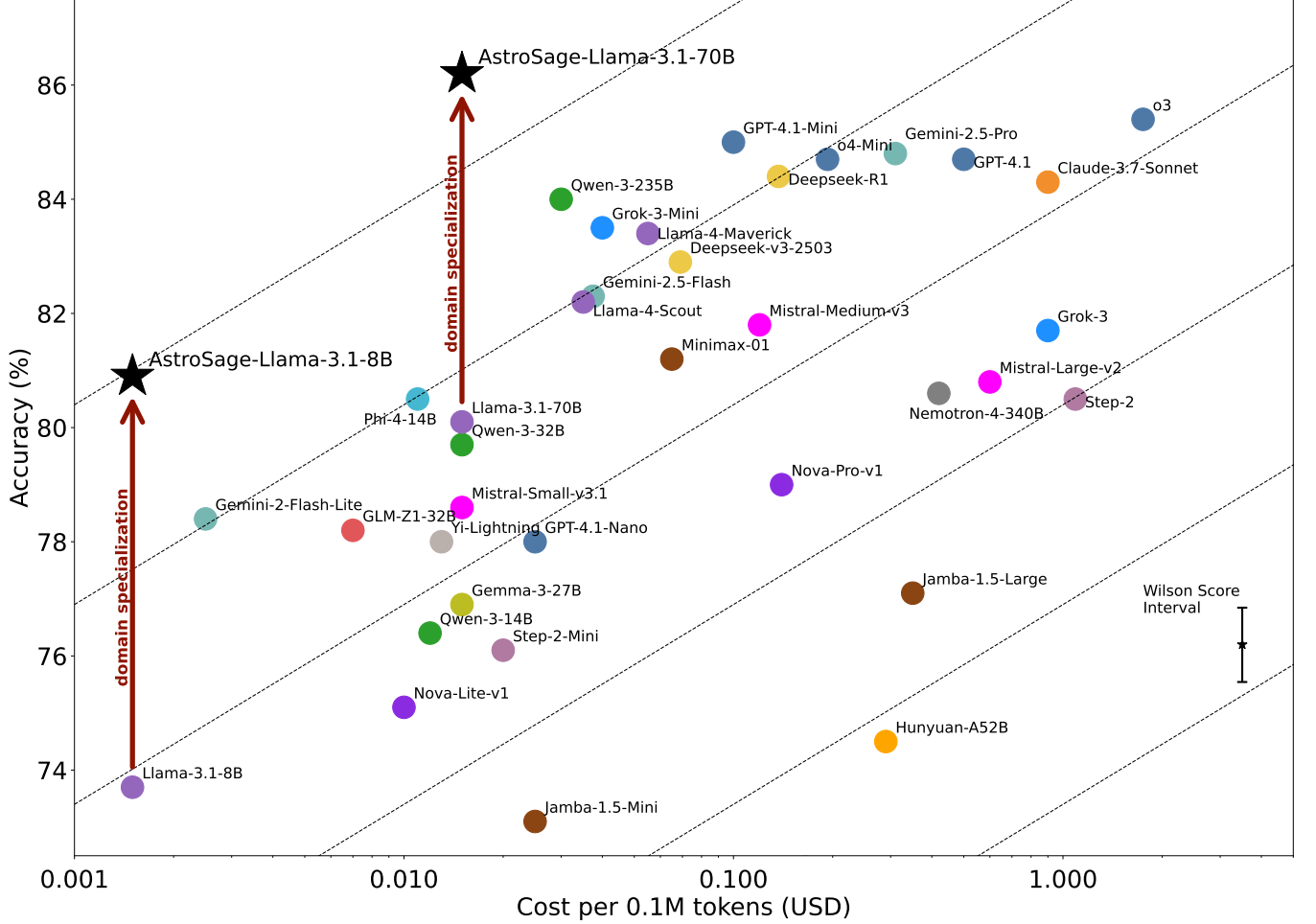


FIG. 3. Performance comparison on the AstroMLab-1 benchmark across 38 LLMs as of May 2025. The x-axis shows cost per 0.1M tokens (USD) on a logarithmic scale, while the y-axis shows accuracy percentage. AstroSage-Llama-3.1-70B achieves 86.2%, outperforming all other models including more expensive proprietary offerings like o3, Claude-3.7-Sonnet, and GPT-4.1. The diagonal dashed lines represent cost-efficiency trade-offs, where a tenfold increase in cost typically corresponds to a 3.5 percentage point improvement in accuracy. The vertical red arrows highlight the effect of domain specialization, showing that both AstroSage models jump approximately two cost-efficiency lines compared to their base models (Llama-3.1-8B and Llama-3.1-70B), representing a roughly 100× improvement in cost-efficiency. The Wilson Score interval in the bottom right shows the typical uncertainty due to the finite number of questions. Figure adapted from Ting *et al.* [1].

than the complex multi-step thought process where reasoning modes typically demonstrate advantages.

We acknowledge this as a limitation of current astronomy benchmarks, which offer limited evaluation of problem-solving capabilities requiring deep reasoning. Nevertheless, since a primary goal of our specialized training is to imbue the model with comprehensive domain knowledge, we consider this benchmark a reasonable test of our model’s capabilities. Future work should develop more complex reasoning-based astronomy benchmarks to better assess the full potential of reasoning-enhanced models in this domain.

The exceptional performance of AstroSage-Llama-3.1-70B on this benchmark supports our central hypothesis: domain specialization, when scaled to larger models, can enable specialized models to surpass even the most ad-

vanced general-purpose commercial models within their domain of expertise.

V. AVAILABILITY

To encourage adoption, further research, and community-driven improvements, AstroSage-Llama-3.1-70B is made freely available under the Llama 3.1 Community License. The full model weights can be accessed and downloaded from our project repository on Hugging Face: <https://huggingface.co/AstroMLab/AstroSage-70B>.

We hope that by providing open access to AstroSage-Llama-3.1-70B, we can accelerate the development and application of advanced AI tools within the astronomical community.

VI. DISCUSSION AND CONCLUSION

The development of AstroSage-Llama-3.1-70B advances specialized language models for astronomy. Building on the foundation established by AstroSage-Llama-3.1-8B, this 70-billion parameter model incorporates a more powerful base architecture, enriched training datasets, refined training methodologies, and explicit reasoning capabilities. In the Ting *et al.* [1] benchmark, AstroSage-Llama-3.1-70B achieved state-of-the-art performance with a score of 86.2%, outperforming all other models at the time of writing, including proprietary models that are orders of magnitude more expensive to operate.

Our results support the central hypothesis of this work: domain specialization, when scaled to larger models, can enable specialized systems to surpass even the most advanced general-purpose commercial models within their domain of expertise. However, it is crucial to note that this superior performance on astronomy-specific tasks does not necessarily translate to general capabilities. In tasks requiring complex mathematical reasoning or general problem-solving, more generalized models likely maintain substantial advantages. Nevertheless, through internal testing and preliminary expert evaluations, we have observed that AstroSage-Llama-3.1-70B demonstrates enhanced domain knowledge recall and the ability to make connections between astronomical concepts compared to general-purpose models. While these qualitative assessments are promising, we acknowledge that more rigorous, large-scale arena-type testing would be necessary to fully validate these observations.

An interesting trend emerged in our evaluation regarding the effectiveness of model distillation and scaling. The performance drop-off between flagship models and their smaller variants appears more pronounced than observed a year ago, particularly in the 30-70B parameter range. This is especially evident in models like Qwen-3, where performance decreases more steeply than the historical 3.5 percentage point per order of magnitude cost relationship would predict. This trend becomes even more pronounced at the 8B parameter level, highlighting the potential importance of specialized training for deploying cost-effective models at scale.

The practical implications of these findings for astronomy are still emerging. While AstroSage-Llama-3.1-70B at 70B parameters represents a substantial computational investment, it may prove valuable for tasks that require both scalability and domain expertise, such as processing and summarizing large volumes of astronomical literature. The open-weight nature of AstroSage-Llama-3.1-70B also enables offline processing of proprietary data from institutions like NASA and NSF, addressing a practical consideration for many research applications.

Looking forward, two key areas warrant further investigation:

1. Development of more comprehensive benchmarks that specifically evaluate reasoning capabilities in astro-

nomical contexts, including problem-solving tasks that more closely resemble real research challenges.

2. Integration of AstroSage-Llama-3.1-70B with astronomy-specific tools and workflows, moving toward more comprehensive AI research assistants that can handle both domain knowledge and practical research tasks.

AstroSage-Llama-3.1-70B advances the integration of AI assistants into astronomical research and education. By making such powerful, specialized tools openly available, we aim to democratize access to cutting-edge AI and accelerate scientific discovery. As astronomical data continues to grow in volume and complexity, tools like AstroSage-Llama-3.1-70B will become increasingly valuable for extracting insights, generating hypotheses, and accelerating the pace of scientific discovery.

We welcome feedback from the astronomical community and invite collaboration to further refine and expand upon this work. The open-source nature of AstroSage-Llama-3.1-70B provides a foundation for community-driven improvements and adaptations to specific research needs within astronomy and related fields.

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